CSE 4345: Big Data Analytics

Week 10, Part 2 — Seminal Research & Case Study

Department of Computer Science & Engineering

Premier University, Chittagong

Semester 2026

Today's Agenda

Seminal Papers Covered

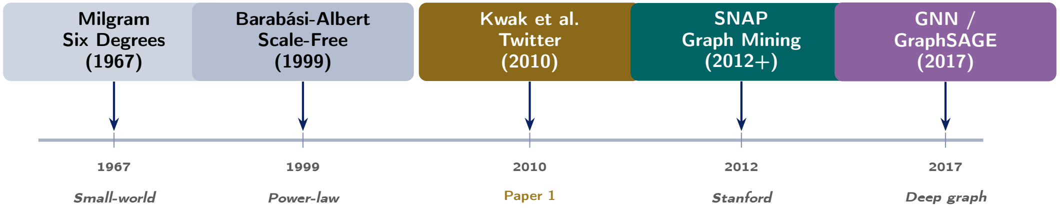
1. Kwak, H., Lee, C., Park, H., & Moon, S. (2010). *What is Twitter, a Social Network or a News Media?* ACM WWW 2010.
2. Taylor, S. J., & Letham, B. (2018). *Forecasting at Scale. The American Statistician, 72*(1).

Case Study

CDC BioSense & ILINet Flu Surveillance (2009–2022): syndromic surveillance of **3,500+ hospitals**, Prophet-based flu nowcasting, and the cautionary tale of **Google Flu Trends** (2008–2015).

Paper 1 — Kwak et al. (2010): Twitter Network Analysis

Research Landscape: Social Network Analysis



Why This Paper Matters for Big Data

Twitter's 2010 graph had **41.7 million users** and **1.47 billion directed following edges**. Crawling, storing, and analysing it required HDFS + MapReduce — the paper is one of the first empirical social network studies at Big Data scale. Its findings shaped every subsequent social media analytics platform, from Twitter's own Trends algorithm to CDC's infodemic surveillance.

Kwak et al. (2010): Publication Context

Full Citation

Kwak, H., Lee, C., Park, H., & Moon, S. (2010). **What is Twitter, a Social Network or a News Media?** In *Proceedings of the 19th International Conference on the World Wide Web (WWW '10)*, pp. 591–600. Raleigh, NC, USA. ACM.

Venue: ACM WWW — Rank A* (CORE), top Web venue
Cited by: >7,000 (Google Scholar, 2024)
Authors: Haewoon Kwak, Changhyun Lee, Hosung Park, Sue Moon — KAIST (Korea Advanced Institute of Science and Technology)

Dataset

Metric	Value
Users crawled	41.7 million
Following edges	1.47 billion
Tweets collected	106 million
Crawl duration	June 2009
Crawl method	BFS via Twitter API
Storage	HDFS cluster
Analysis	MapReduce (Hadoop)

The Central Question

Is Twitter a social network (mutual

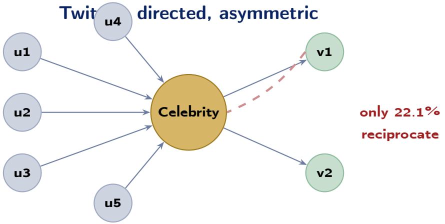
Infrastructure Note

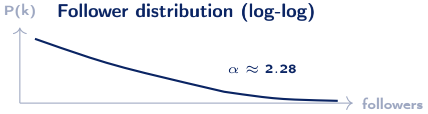
Collecting 1.47 billion edges required

Technical Contribution 1 — Network Topology Findings

Key Structural Results

1. **Highly asymmetric**: 77.9% of users have a follower who does not follow them back. Compare to Facebook (mutual friendship required): Twitter is a *directed* graph, not undirected.
2. **Power-law degree distribution**: follower counts follow $P(k) \propto k^{-\alpha}$ with $\alpha \approx 2.276$ for in-degree (followers). The top 0.05% of users (celebrities) have >1 million followers.
3. **Small world**: average shortest path length = **4.12 hops** across 41.7M users — slightly less than Milgram's "six degrees."
4. **Reciprocity rate**: only 22.1% of following relationships are mutual — far lower than Facebook (≈100%).





Technical Contribution 2 — Twitter as News Medium

Trending Topics: News, Not Social Chatter

Kwak et al. analysed 106M tweets to characterise **Trending Topics**:

- **85%** of trending topics were *headline news* (breaking news, political events, entertainment releases) — not personal conversations
- Only 15% were social memes or conversational trends
- The median duration of a trending topic: **17.5 hours** — the lifespan of a news cycle
- A breaking news topic reaches **peak tweet rate** within **15 minutes** of the event

Conclusion: Twitter functions as a **real-time news wire**, not a social network. Its information diffusion speed rivals (and sometimes beats) professional

Impact on Public Health Surveillance

The Twitter-as-news-media finding directly enabled **social media epidemiology**:

- If Twitter reflects real-world events within minutes, flu-related tweets should correlate with actual flu incidence
- Culotta (2010) and Signorini et al. (2011) used Twitter keyword streams to estimate influenza rates *two weeks ahead* of CDC official reports
- CDC now monitors **Twitter/X for infodemic signals**: vaccine misinformation, outbreak rumours, mental health crisis language
- The paper's directed-graph model of

Paper 2 — Taylor & Letham (2018): Prophet

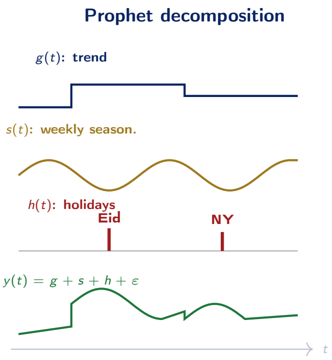
Taylor & Letham (2018): Publication Context & Model

Full Citation

Taylor, S. J., & Letham, B. (2018). **Forecasting at Scale.** *The American Statistician, 72*(1), 37–45. (First published online 2017.)

Venue: The American Statistician (ASA, peer-reviewed)
Cited by: >1,500 (Google Scholar, 2024)
Authors: Sean J. Taylor & Benjamin Letham (Facebook Core Data Science team)

Code: Apache-licensed at `github.com/facebook/prophet`; available for Python and R



The Decomposable Model

Prophet: Changepoints, Seasonality & Scale

Automatic Changepoint Detection

The trend $g(t)$ is **piecewise linear**:

$$g(t) = (k + \mathrm{a}(t)^\top \boldsymbol{\delta})\, t + (m + \mathrm{a}(t)^\top \gamma)$$

where $\boldsymbol{\delta}$ are slope adjustments at automatically detected **changepoints** (structural breaks in trend).

Sparse prior on $\boldsymbol{\delta}$: most changepoints have $\delta_j \approx 0$; a few are large. This avoids overfitting while capturing genuine trend shifts.

Fourier seasonality:

$$s(t) = \sum_{n=1}^{N} \left(a_n \cos\tfrac{2\pi nt}{P} + b_n \sin\tfrac{2\pi nt}{P}\right)$$

$P = 365.25$ (annual), $P = 7$ (weekly); N controls

Designed for Analysts, Not Statisticians

Prophet's key design goal was **scalable human-in-the-loop forecasting** at Facebook:

- Facebook needed forecasts for **thousands of business metrics** (DAU, ad impressions, revenue) updated daily
- Domain analysts (not statisticians) needed to **tune** forecasts — add known holidays, adjust trend flexibility, handle outliers
- **Interpretable parameters**: analysts understand "weekly seasonality" and "holiday dip" without understanding ARIMA lag orders
- **Robust to missing data and outliers**: the additive structure absorbs spikes without corrupting the trend estimate

Impact Today — Kwak (2010) & Taylor/Letham (2018)

Kwak et al. — Downstream Impact

- **Twitter Trending Topics algorithm**: the paper's finding that trending topics are news-driven shaped Twitter's later algorithm redesigns
- **Social media epidemiology**: CDC, WHO, and academic groups built flu, COVID, and mental health surveillance on the Twitter graph model
- **Fake news research**: the directed-cascade model of retweets is the reference network structure for misinformation propagation studies (Vosoughi 2018)
- **Influence maximisation** (Kempe et al. 2003 + Kwak findings): seeding a campaign at high-PageRank nodes vs. high-follower nodes

Prophet — Downstream Impact

- **Industry standard for business forecasting**: adopted by Airbnb, Uber, Walmart, and dozens of unicorn-stage startups for demand and revenue forecasting
- **Healthcare**: hospital admissions forecasting; ICU capacity planning during COVID-19 waves
- **Energy sector**: electricity demand forecasting with Prophet + temperature covariates
- **Spark vectorisation**: the `applyInPandas` + Prophet pattern (Week 10, Part 1) is now a canonical PySpark design pattern in production
- **NeuralProphet** (2021): replaces Fourier

Case Study — CDC Flu Surveillance & the Google Flu Lesson

CDC Disease Surveillance Infrastructure

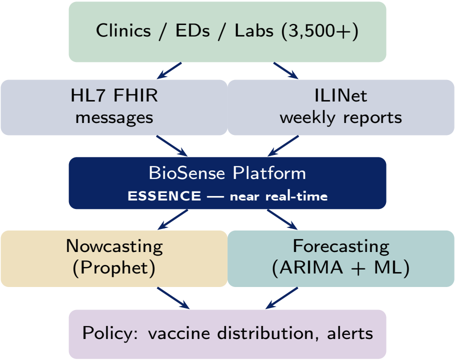
The CDC ILINet and BioSense Systems

ILINet (Influenza-Like Illness Surveillance Network):

- >3,000 sentinel physicians in all 50 US states report weekly counts of ILI patients (*fever + cough or sore throat*)
- Data flows: clinic → state health dept → CDC weekly report
- **Lag**: 1–2 weeks behind real time
- Used to issue **flu season start/end** and estimate excess mortality

BioSense Platform:

- Real-time electronic health data from **3,500+ emergency departments**
- Syndromes (flu-like, gastrointestinal, neurological) automatically extracted from visit



Google Flu Trends: A Big Data Cautionary Tale

The Promise (2008–2011)

Google Flu Trends (GFT), launched 2008 (Ginsberg et al., *Nature* 2009):

- Correlated 45 flu-related search queries with ILINet data to **predict flu incidence** *two weeks before* CDC official reports
- Initial accuracy: $r^2 = 0.97$ on 2003–2008 data
- Cited as the canonical example of **Big Data surpassing traditional surveillance**

The Failure (2012–2015)

GFT over-predicted flu rates by $2\times$ in the 2012–2013 season. Lazer et al. (*Science* 2014) diagnosed three causes:

Prophet-Based Nowcasting: A Better Approach

Post-GFT failure, CDC and academic groups rebuilt flu forecasting using **Prophet + BioSense**:

- **BioSense ED visits** as the primary signal (direct medical observation, not proxy search data)
- **Prophet** models weekly and annual seasonality; detects changepoints at flu season start
- **Twitter signals** (Kwak paper) added as an *exogenous covariate*, not a replacement for surveillance data
- **Ensemble**: CDC FluSight challenge

Lessons Learned & Discussion Questions

Key Lessons from the CDC/GFT Case

1. **Proxy data amplifies noise**: search queries capture *health anxiety*, not health incidence; proxy signals need causal validation before policy use
2. **Algorithm transparency is a public health issue**: GFT failed because Google's internal autocomplete changes (opaque to CDC) shifted the feature distribution
3. **Decomposability improves trust**: Prophet's $g + s + h$ structure lets epidemiologists inspect each component — unlike black-box deep learning forecasters
4. **Ensemble beats any single model**: CDC FluSight's top performers (2013–2022) are always ensemble methods; Prophet is a strong

Discussion Questions (CLO4)

1. Kwak et al. found Twitter's average shortest path is 4.12 hops over 41.7M users. If a misinformation tweet is retweeted by a high-PageRank account, estimate how many users it reaches within **3 hops** assuming an average out-degree of 45. What graph property makes this estimate an **overestimate**? *[CLO4 — Apply]*
2. A Prophet model for weekly hospital admissions has strong annual and weekly seasonality but misses a sudden COVID-19 wave onset. Identify which Prophet component should detect this changepoint and explain what prior

Live Exercise

Live Exercise — Prophet Flu Forecasting in Python

Task (25 minutes, pairs)

Fit a **Prophet** model to synthetic weekly flu incidence data and produce a 12-week forecast.

Goals:

1. Generate 3 years of synthetic weekly ILI data with annual seasonality + noise.
2. Fit Prophet with yearly and weekly seasonality.
3. Plot the forecast with 95% uncertainty intervals.
4. Add a **holiday** for "Eid" (spike in healthcare visits) and observe the effect.
5. Change `changepoint_prior_scale` from 0.05 to 0.5 and compare trend flexibility.

Install

Python Skeleton

```
import pandas as pd
import numpy as np
from prophet import Prophet
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
dates = pd.date_range("2021-01-04",
                       periods=156, freq="W")
# Annual flu seasonality + noise
ili = (5 + 3*np.sin(
         2*np.pi*np.arange(156)/52
         - np.pi/2)
       + rng.normal(0, 0.5, 156)).clip(0)

df = pd.DataFrame({"ds": dates, "y": ili})

# Optional: add Eid holiday
holidays = pd.DataFrame({
  "holiday":  "Eid",
  "ds":       pd.to_datetime([
                "2021-05-13","2022-05-03",
                "2023-04-22"]),
  "lower_window": 0,
```

References

Seminal Papers

- Kwak, H., Lee, C., Park, H., & Moon, S. (2010). What is Twitter, a social network or a news media? *Proc. ACM WWW*, 591–600.
- Taylor, S. J., & Letham, B. (2018). Forecasting at scale. *The American Statistician, 72*(1), 37–45.

Textbooks [SA], [TE], [LRU]

- **[SA]** Acharya, S., & Chellappan, S. (2015). *Big Data and Analytics* (Ch. 11 — Social Media Analytics). Wiley.
- **[TE]** Erl, T., Khattak, W., & Buhler, P. (2016). *Big Data Fundamentals* (Ch. 12 — Time Series and Health Data). Prentice Hall.
- **[LRU]** Leskovec, J., Rajaraman, A., & Ullman, J. D. (2020). *Mining of Massive Datasets*, 3rd ed. (Ch. 10 – Social Network Graphs). Cambridge UP.

Case Study Sources

- Lazer, D., et al. (2014). The parable of Google Flu: Traps in big data analysis. *Science*

Closing — Week 10, Part 2

Key Takeaways

- Kwak et al. (2010) proved that Twitter is a **news medium, not a social network**: this single finding unlocked social media as a real-time epidemiological surveillance tool and a misinformation propagation model
- Prophet's **decomposable, interpretable structure** ($g + s + h$) is why it succeeded where black-box models failed in public health: domain experts can audit and adjust each component
- Google Flu Trends is the canonical **Big Data hubris story**: high in-sample r^2 does not guarantee out-of-sample generalisation when the data-generating process is non-stationary
- The combination of **BioSense (direct signal)** + **Twitter (leading indicator)** + **Prophet**

Quote of the Week

"Big data hubris is the often implicit assumption that big data are a substitute for, rather than a supplement to, traditional data collection and analysis."

— Lazer et al., *Science* (2014)

Part 1 Preview — Week 11

Lakehouse, Data Mesh, MLOps & Edge Analytics

- Delta Lake / Apache Iceberg ACID semantics